

Synchronization of Reaction–Diffusion Neural Networks With Dirichlet Boundary Conditions and Infinite Delays

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Abstract—This paper is concerned with synchronization for a class of reaction–diffusion neural networks with Dirichlet boundary conditions and infinite discrete time-varying delays. By utilizing theories of partial differential equations, Green’s formula, inequality techniques, and the concept of comparison, algebraic criteria are presented to guarantee master-slave synchronization of the underlying reaction–diffusion neural networks via a designed controller. Additionally, sufficient conditions on exponential synchronization of reaction–diffusion neural networks with finite time-varying delays are established. The proposed criteria herein enhance and generalize some published ones. Three numerical examples are presented to substantiate the validity and merits of the obtained theoretical results.

Index Terms—Comparison, Dirichlet boundary condition, infinite delay, reaction–diffusion neural networks, synchronization.

I. INTRODUCTION

DURING the last three decades, neural networks have received considerable attention from numerous fields of science and technology due to their significant applications in associative memories, signal processing, pattern recognition, and optimization problems [1]–[3]. These applications are tightly associated with the dynamical properties of neural networks. Consequently, in the design and implementation of neural networks, dynamical analysis is indispensable [4]–[6].

Time delays do exist in real-world systems owing to the finite information exchanging speeds between different

units, which may cause instability, oscillation, or divergence [7]–[13]. When a neural network model is designed, time delays should therefore be taken into consideration. Based on this fact, a great number of outstanding achievements on dynamical behavior analysis of various delayed neural networks have recently been published in [14]–[45], to name a few.

Many previous accomplishments on dynamical analysis of neural networks mainly focus on the models described by ordinary differential equations or functional differential equations, please refer to [2], [4], [5], [14], [15], [18], [30], [39], [42]–[45], and references therein. In these models, state variables of neurons only vary with time. It should be pointed out that diffusion effects need to be considered in neural networks and electric circuits when electrons are moving in a nonuniform electromagnetic field [34], [36], [46], [47]. Namely, state variables of neurons depend not only on time variable but also on space variable. In this case, neural network models should be described by partial differential equations (PDEs) through introducing reaction–diffusion terms. Actually, there exist plenty of reaction–diffusion phenomena in the areas of biology and chemistry. For example, in [48], a diffusion term $\sum_{k=1}^m (\partial/\partial x_k)(D_i(\partial N_i/\partial x_k))$ is added into the famous Lotka–Volterra equations as follows:

$$\frac{\partial N_i}{\partial t} = N_i \left(r_i + \sum_{j=1}^n a_{ij} N_j \right) + \sum_{k=1}^m \frac{\partial}{\partial x_k} \left(D_i \frac{\partial N_i}{\partial x_k} \right) \quad (1)$$

in which $i = 1, 2, \dots, n$, the positive integer n means that there are n species in the population dynamics, N_i denotes the population level of the species i , and D_i corresponds to the measure of dispersal rate. Sufficient conditions ensuring global stability of (1) are developed therein.

Nowadays, reaction–diffusion neural networks have stirred much research interest owing to their prominent values in image processing, pattern recognition, and shortest path problem [49]. Numerous elegant studies on dynamical behavior analysis of reaction–diffusion neural networks have been accumulated, please refer to [23], [28], [29], [33]–[37], [46], [49]–[62], and relevant references therein.

Synchronization, as a special case of dynamical behaviors, is a very hot issue in many scientific disciplines, because it can be applied to combinatorial optimization, image processing, and secure communication [63]. In recent years,

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many efforts have been dedicated to investigating synchronization of reaction–diffusion neural networks with time delays [28], [37], [46], [49], [54]–[62]. It should be emphasized that discrete time delays in reaction–diffusion neural network models [28], [29], [33], [34], [36], [37], [49], [51]–[53], [58]–[62] are assumed to have upper bounds.

Information exchange between different neurons is a time-varying activity. In the fields of dynamical analysis for reaction–diffusion neural networks with discrete delays, the discrete delays are generally assumed to be bounded. It means that current states of a neuron only depend on a part of the history. In fact, present behaviors of a neuron depend upon all its historical information. As a result, discrete delays in neural networks should be hypothesized to be time-varying and infinite, which can reveal the characteristics of neurons in human brains in a more realistic way.

To the best of the authors' knowledge, there are few published studies concerning master-slave synchronization of reaction–diffusion neural networks with Dirichlet boundary conditions and infinite discrete time-varying delays, because of the difficulties stemmed from the partial differential operators and infinite delays. It is an important and challenging problem both in theoretical interest and practical applications.

Based on the aforementioned discussion, in this paper, we intend to investigate master-slave synchronization for a general class of reaction–diffusion neural networks with Dirichlet boundary conditions and infinite discrete time-varying delays. With the help of theories of PDEs, Green's formula, inequality techniques, and comparison method, algebraic criteria in terms of p -norm ($p \geq 2$) are provided to ensure asymptotic and exponential stability of error neural networks. Then, asymptotic and exponential synchronization of master-slave neural networks can be realized. Meanwhile, sufficient conditions are presented to guarantee exponential synchronization of reaction–diffusion neural networks with finite time-varying delays. Especially, an M -matrix-based criterion is obtained, which improves and generalizes the ones in [4], [5], [15], and [52]. Simulation examples are followed, which illustrate that the achieved theoretical results in this paper are less conservative than the ones in [34] and [36].

The remaining portion of this paper is organized as follows. In Section II, reaction–diffusion neural network models and definitions of synchronization as well as some important lemmas are given. The main results are presented in Section III. Three numerical examples are provided in Section IV. Conclusions are collected in Section V.

Notations: Throughout this paper, the notations are standard. $\mathbb{R}^n$ represents the n -dimensional Euclidean space, and $\mathbb{R}$ is the set of real numbers. $\hat{m}$ is the set $\{1, 2, \dots, m\}$ where m is a positive integer. A^T represents the transpose of A . $\Omega = \{x | x = [x_1, x_2, \dots, x_m]^T, |x_s| < q_s, s \in \hat{m}\}$ is a bounded compact set with smooth boundary $\partial\Omega$ and $\text{mes}\Omega > 0$ where $\text{mes}\Omega$ is the measure of Ω , and $q_s, s \in \hat{m}$ are given scalars. $H = [h_{ij}]_{n \times n}$ means that H is a real square matrix with elements h_{ij} , $i, j \in \hat{n}$. $B = \text{diag}\{b_1, b_2, \dots, b_n\}$ denotes a diagonal matrix. Vector $\vec{r} \gg 0$ means that every element of $\vec{r}$ is bigger than 0. $\mathbb{C}^0$ denotes the set of continuous functions, and $\mathbb{C}^1$ means that

the first derivative does exist and is continuous. $\mathbb{B} \triangleq \mathbb{C}^0[\Omega \times (-\infty, t_0] \rightarrow \mathbb{R}^n]$ represents the Banach space of all continuous functions $\psi(x, s) : \Omega \times (-\infty, t_0] \rightarrow \mathbb{R}^n$ equipped with p -norm defined by $\|\psi(x, s)\|_p = \{\int_{\Omega} \sum_{i=1}^n |\psi_i(x, s)|^p dx\}^{(1/p)}$. $\mathbb{D}^+ \chi(t)$ corresponds to the upper right Dini derivative and is defined as $\mathbb{D}^+ \chi(t) = \limsup_{s \rightarrow 0^+} [(\chi(t+s) - \chi(t))/s]$.

II. PROBLEM DESCRIPTION

Consider a class of reaction–diffusion neural networks with Dirichlet boundary conditions and time-varying delays

$$\begin{aligned} \frac{\partial u_i(x, t)}{\partial t} = & \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial u_i(x, t)}{\partial x_k} \right] - b_i u_i(x, t) \\ & + \sum_{j=1}^n c_{ij} \tilde{f}_j(u_j(x, t)) \\ & + \sum_{j=1}^n h_{ij} \tilde{g}_j(u_j(x, t - d_{ij}(t))) + I_i, \quad i \in \hat{n} \quad (2) \end{aligned}$$

in which $x = [x_1, x_2, \dots, x_m]^T \in \Omega \subseteq \mathbb{R}^m$, $u_i(x, t)$ represents the state of the i th neuron in space x and at time t , $a_{ik} \geq 0$ denotes the transmission diffusion coefficient along the i th neuron, $\sum_{k=1}^m (\partial/\partial x_k)[a_{ik}(\partial u_i(x, t)/\partial x_k)]$ is a reaction–diffusion term, $b_i > 0$ stands for the velocity with which the i th neuron will reset its potential to the resting state in isolation when disconnected from the neural networks and external inputs, c_{ij} and h_{ij} are connection weight coefficient and delayed connection weight coefficient, respectively, $d_{ij}(t)$ is the transmission delay, $\tilde{f}_j(\cdot)$ and $\tilde{g}_j(\cdot)$ correspond to activation functions of the j th neuron without and with time delays, respectively, and I_i means an external input.

Associated with neural networks (2), for any initial time $t_0 \geq 0$, initial value condition and boundary value condition are given by

$$u_i(x, s) = \psi_i(x, s), \quad (x, s) \in \Omega \times (-\infty, t_0], \quad i \in \hat{n} \quad (3)$$

$$u_i(x, t) = 0, \quad (x, t) \in \partial\Omega \times (-\infty, +\infty), \quad i \in \hat{n}. \quad (4)$$

$\psi(x, s) = [\psi_1(x, s), \psi_2(x, s), \dots, \psi_n(x, s)]^T \in \mathbb{B}$ is a bounded and continuous vector-valued function.

Remark 1: Compared with the reaction–diffusion neural network models in [36], [52], [53], and [62], neural networks (2) are more general owing to the different activation functions $\tilde{f}_j(\cdot)$ and $\tilde{g}_j(\cdot)$ at time t and $t - d_{ij}(t)$, respectively. On the other hand, let $a_{ik} = 0$, $i \in \hat{n}$, $k \in \hat{m}$, then neural networks (2) reduce to the following neural network models:

$$\begin{aligned} \frac{du_i(t)}{dt} = & -b_i u_i(t) + \sum_{j=1}^n c_{ij} \tilde{f}_j(u_j(t)) \\ & + \sum_{j=1}^n h_{ij} \tilde{g}_j(u_j(t - d_{ij}(t))) + I_i, \quad i \in \hat{n} \quad (5) \end{aligned}$$

which include some neural network models as special cases. For example, let $\tilde{f}_j$ be the sigmoid function and $\tilde{g}_j = 0$, $j \in \hat{n}$, then neural networks (5) reduce to the continuous-time

Hopfield neural network models. Let $\tilde{f}_j(u) = (1/2)(|u + 1| - |u - 1|)$ and $\tilde{g}_j(u) = 0$ or $\tilde{g}_j(u) = \tilde{f}_j(u)$, $j \in \hat{n}$, then neural networks (5) correspond to the cellular neural networks without or with time delays. Let $\tilde{f}_j = 0$, $\tilde{g}_j(u_j) = \tanh(\lambda_j u_j)$, $j \in \hat{n}$, and $H = [h_{ij}]_{n \times n} = \begin{bmatrix} 0 & H_1 \\ H_2 & 0 \end{bmatrix}$, then neural networks (5) turn into the bidirectional associative memory networks. Theoretically, the obtained criteria in this paper can be deduced to the ones of the above-mentioned neural network models.

To investigate master-slave synchronization, in what follows, two assumptions are given.

Assumption 1: Activation functions $\tilde{f}_i(\cdot)$ and $\tilde{g}_i(\cdot)$, $i \in \hat{n}$ in neural networks (2) are bounded and satisfy the Lipschitz condition, that is, there exist constants $l_i, \tilde{l}_i > 0$ such that

$$\begin{aligned} |\tilde{f}_i(u_1) - \tilde{f}_i(u_2)| &\leq l_i |u_1 - u_2|, \quad \forall u_1, u_2 \in \mathbb{R} \\ |\tilde{g}_i(u_1) - \tilde{g}_i(u_2)| &\leq \tilde{l}_i |u_1 - u_2|, \quad \forall u_1, u_2 \in \mathbb{R}. \end{aligned} \quad (6)$$

Assumption 2: Time-varying delays $d_{ij}(t) \geq 0$, $i, j \in \hat{n}$ in neural networks (2) are unbounded and continuous.

Remark 2: In [33], [34], [36], and [53], activation functions need to be monotonically nondecreasing. This restriction is removed in this paper. Activation functions $\tilde{f}_i(\cdot)$ and $\tilde{g}_i(\cdot)$, $i \in \hat{n}$ in neural networks (2) can be nonmonotonic.

Remark 3: Discrete time delays in [28], [29], [33], [34], [36], [37], [49], [51]–[53], and [58]–[62] are assumed to have upper bounds. In neural networks (2), discrete time-varying delays $d_{ij}(t)$, $i, j \in \hat{n}$ can be unbounded.

In this paper, neural networks (2) are referred to as the master neural networks, and slave neural networks are described by the following PDEs:

$$\begin{aligned} \frac{\partial v_i(x, t)}{\partial t} &= \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial v_i(x, t)}{\partial x_k} \right] - b_i v_i(x, t) \\ &+ \sum_{j=1}^n c_{ij} \tilde{f}_j(v_j(x, t)) + \sum_{j=1}^n h_{ij} \tilde{g}_j(v_j(x, t - d_{ij}(t))) \\ &+ I_i + \Delta_i(x, t), \quad i \in \hat{n} \end{aligned} \quad (7)$$

where $\Delta_i(x, t)$ is a designed controller by

$$\Delta_i(x, t) = \sum_{j=1}^n \delta_{ij} [v_j(x, t) - u_j(x, t)] \quad (8)$$

where δ_{ij} is a constant and $\delta_{ii} < 0$. Other notations have same definitions as those in neural networks (2).

Associated with neural networks (7), initial value condition and boundary value condition are given as follows:

$$v_i(x, s) = \varphi_i(x, s), \quad (x, s) \in \Omega \times (-\infty, t_0], \quad i \in \hat{n} \quad (9)$$

$$v_i(x, t) = 0, \quad (x, t) \in \partial\Omega \times (-\infty, +\infty), \quad i \in \hat{n} \quad (10)$$

where $\varphi(x, s) = [\varphi_1(x, s), \varphi_2(x, s), \dots, \varphi_n(x, s)]^T \in \mathbb{B}$ is bounded and continuous.

The main purpose of this paper is to design an appropriate controller (8) such that master neural networks (2) and slave neural networks (7) can realize synchronization. To do this,

define the synchronization error in the following form:

$$e_i(x, t) = \frac{1}{r_i} [v_i(x, t) - u_i(x, t)] \quad (11)$$

$r_i > 0$, $i \in \hat{n}$. In view of neural networks (2) and (7), error neural networks are obtained as follows:

$$\begin{aligned} \frac{\partial e_i(x, t)}{\partial t} &= \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right] - b_i e_i(x, t) \\ &+ \sum_{j=1}^n \frac{c_{ij}}{r_i} f_j(v_j(x, t), u_j(x, t)) \\ &+ \sum_{j=1}^n \frac{h_{ij}}{r_i} g_j(v_j(x, t - d_{ij}(t)), u_j(x, t - d_{ij}(t))) \\ &+ \sum_{j=1}^n \frac{r_j}{r_i} \delta_{ij} e_j(x, t), \quad i \in \hat{n} \end{aligned} \quad (12)$$

in which $f_j(v_j(x, t), u_j(x, t)) = \tilde{f}_j(v_j(x, t)) - \tilde{f}_j(u_j(x, t))$ and $g_j(v_j(x, t - d_{ij}(t)), u_j(x, t - d_{ij}(t))) = \tilde{g}_j(v_j(x, t - d_{ij}(t))) - \tilde{g}_j(u_j(x, t - d_{ij}(t)))$, $j \in \hat{n}$. Other notations have similar definitions as those in neural networks (2) and (7). Meanwhile, combining with the Lipschitz conditions in Assumption 1, for $\forall i \in \hat{n}$, $u_3, u_4 \in \mathbb{R}$

$$|f_i(u_3, u_4)| \leq l_i |u_3 - u_4|, \quad |g_i(u_3, u_4)| \leq \tilde{l}_i |u_3 - u_4|. \quad (13)$$

Corresponding to neural networks (2) and (7), initial value condition and boundary value condition of neural networks (12) are illustrated by

$$e_i(x, s) = \frac{\varphi_i(x, s) - \psi_i(x, s)}{r_i}, \quad (x, s) \in \Omega \times (-\infty, t_0] \quad (14)$$

$$e_i(x, t) = 0, \quad (x, t) \in \partial\Omega \times (-\infty, +\infty) \quad (15)$$

$i \in \hat{n}$. $[(\varphi_1(x, s) - \psi_1(x, s))/r_1], [(\varphi_2(x, s) - \psi_2(x, s))/r_2], \dots, [(\varphi_n(x, s) - \psi_n(x, s))/r_n]^T \in \mathbb{B}$ is bounded and continuous. Besides, as to the vector $e(x, t) = [e_1(x, t), e_2(x, t), \dots, e_n(x, t)]^T : \Omega \times [t_0, +\infty) \rightarrow \mathbb{R}^n$, p -norm ($p \geq 2$) of $e(x, t)$ is defined as

$$\|e(x, t)\|_p = \left\{ \int_{\Omega} \sum_{i=1}^n |e_i(x, t)|^p dx \right\}^{\frac{1}{p}}. \quad (16)$$

Before proceeding, some definitions and lemmas are given which have important roles in the proof.

Definition 1: Master neural networks (2) and slave neural networks (7) are globally asymptotically synchronized under the designed controller (8) if error neural networks (12) are locally stable in the sense of Lyapunov and are globally attractive. Furthermore, neural networks (2) and (7) are globally exponentially synchronized under the designed controller (8) if there exist two constants $\varepsilon > 0$ and $M > 0$ such that

$$\|e(x, t)\|^p \leq M \sup_{-\infty \leq s \leq t_0} \|\varphi - \psi\|^p e^{-\varepsilon(t-t_0)}, \quad t \geq t_0. \quad (17)$$

Definition 2 [64]: A square matrix $\mathbb{S} = [s_{ij}]_{n \times n}$ is an M -matrix if $\mathbb{S}$ can be expressed in the form of $\mathbb{S} = s\mathbb{I} - G$ with some $G \geq 0$ and $s > \rho(G)$, where $\mathbb{I}$ is the identity $n \times n$ matrix and $\rho(G)$ is the spectral radius of G .

Lemma 1 (Young's Inequality): Let p and q be positive real numbers satisfying $(1/p) + (1/q) = 1$. Then for any nonnegative real numbers a and b

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \quad (18)$$

The equality holds if and only if $a^p = b^q$.

Lemma 2 [57]: Let $p \geq 2$ be a positive integer, $q_k, k \in \hat{n}$ be positive constants, $\Omega = \{x|x = [x_1, x_2, \dots, x_m]^T, |x_k| < q_k\}$ be a cube and $h(x)$ be a real-valued function belonging to $\mathbb{C}^1(\Omega)$ which vanishes on the boundary $\partial\Omega$ of Ω , i.e., $h(x)|_{x \in \partial\Omega} = 0$. Then for any $k \in \hat{n}$

$$\int_{\Omega} |h(x)|^p dx \leq \frac{p^2 q_k^2}{4} \int_{\Omega} |h(x)|^{p-2} \left[\frac{\partial h(x)}{\partial x_k} \right]^2 dx. \quad (19)$$

Lemma 3 [17], [65]: Consider constants $a_i \geq 0, s_i > 0$ ($i = 1, 2, \dots, n+1$), then the following inequality holds:

$$\left\{ \prod_{i=1}^{n+1} a_i^{s_i} \right\}^{\frac{1}{S_{n+1}}} \leq \left\{ \sum_{i=1}^{n+1} s_i a_i^p \right\}^{\frac{1}{p}} \frac{1}{\sqrt[p]{S_{n+1}}} \quad (20)$$

where $p > 0$ and $S_{n+1} = \sum_{i=1}^{n+1} s_i$. Especially, let $S_{n+1} = p = \sum_{i=1}^n s_i + 1$, then (20) reduces to

$$\left\{ \prod_{i=1}^n a_i^{s_i} \right\} a_{n+1} \leq \frac{1}{p} \left\{ \sum_{i=1}^n s_i a_i^p + a_{n+1}^p \right\}. \quad (21)$$

Lemma 4 [64]: Let $\mathbb{S} = [s_{ij}]_{n \times n}$ be a square matrix with nonpositive off-diagonal elements. Then the following statements are equivalent.

- 1) $\mathbb{S} = [s_{ij}]_{n \times n}$ is an M -matrix.
- 2) The real parts of all eigenvalues of $\mathbb{S}$ are positive.
- 3) There exists a vector $\bar{\mathbf{r}} \gg 0$ such that $\mathbb{S}\bar{\mathbf{r}} \gg 0$.

III. MAIN RESULTS

To study master-slave synchronization of delayed reaction-diffusion neural networks (2) and (7) via the designed controller (8), we turn to consider stability of delayed neural networks (12) with initial value condition (14) and boundary value condition (15). Now the main results are given.

Theorem 1: Given constants $p \geq 2, q_s, s \in \hat{n}$ and $l_i, \tilde{l}_i, i \in \hat{n}$. Suppose that Assumptions 1 and 2 are satisfied. Neural networks (12) are globally asymptotically stable, namely, master neural networks (2) and slave neural networks (7) are globally asymptotically synchronized under the designed controller (8) in the sense of Definition 1, if there exist functions $y_i(t) \in \mathbb{C}^0(\mathbb{R}, \mathbb{R})$ with $y_i(s) \leq 0, \forall s \leq t_0$, and $\lim_{t \rightarrow +\infty} \int_{t_0}^t y_i(s) ds = -\infty, i \in \hat{n}$, positive integers $\Gamma_1, \Gamma_2, \Gamma_3$, positive constants $\mu_k, k \in \hat{\Gamma}_1, v_k, k \in \hat{\Gamma}_2, \omega_k, k \in \hat{\Gamma}_3, r_k, k \in \hat{n}$, negative constants $\delta_{ii}, i \in \hat{n}$ and any real constants $\delta_{ij}, i \neq j, \alpha_{ij}, \alpha_{ij}^*, \beta_{ij}, \beta_{ij}^*, \theta_{ij}, \theta_{ij}^*, \varrho_{ij}, \varrho_{ij}^*, \sigma_{ij}, \sigma_{ij}^*, i, j \in \hat{n}$ such

that for any $t \geq t_0 > 0$

$$\begin{aligned} & \left[- \sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} + p(\delta_{ii} - b_i + |c_{ii}|l_i) \right. \\ & + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_1} \frac{r_j}{r_i} \mu_k |c_{ij}|^{\frac{pq_{ij}}{\mu_k}} l_j^{\frac{p\beta_{ij}}{\mu_k}} + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_2} \frac{r_j}{r_i} v_k |\delta_{ij}|^{\frac{p\theta_{ij}}{v_k}} \\ & + \sum_{j=1}^n \sum_{k=1}^{\Gamma_3} \frac{r_j}{r_i} \omega_k |h_{ij}|^{\frac{pq_{ij}}{\omega_k}} \tilde{l}_j^{\frac{p\sigma_{ij}}{\omega_k}} - y_i(t) \left. \right] \exp \left\{ \int_{t_0}^t y_i(s) ds \right\} \\ & + \sum_{j=1, j \neq i}^n \frac{r_j}{r_i} \left[|c_{ij}|^{p\alpha_{ij}^*} l_j^{p\beta_{ij}^*} + |\delta_{ij}|^{p\theta_{ij}^*} \right] \exp \left\{ \int_{t_0}^t y_j(s) ds \right\} \\ & + \sum_{j=1}^n \frac{r_j}{r_i} |h_{ij}|^{p\varrho_{ij}^*} \tilde{l}_j^{p\sigma_{ij}^*} \exp \left\{ \int_{t_0}^{t-d_{ij}(t)} y_j(s) ds \right\} < 0 \end{aligned} \quad (22)$$

in which

$$\begin{aligned} \Gamma_1 \alpha_{ij} + \alpha_{ij}^* &= 1, \quad \Gamma_1 \beta_{ij} + \beta_{ij}^* = 1, \quad \Gamma_2 \theta_{ij} + \theta_{ij}^* = 1 \\ \Gamma_3 \varrho_{ij} + \varrho_{ij}^* &= 1, \quad \Gamma_3 \sigma_{ij} + \sigma_{ij}^* = 1 \\ \sum_{k=1}^{\Gamma_1} \mu_k &= \sum_{k=1}^{\Gamma_2} v_k = \sum_{k=1}^{\Gamma_3} \omega_k = p-1. \end{aligned} \quad (23)$$

Furthermore, neural networks (12) are globally exponentially stable, that is, neural networks (2) and (7) are globally exponentially synchronized via the designed controller (8) in the sense of Definition 1, if there exist constants $M > 0$ and $\epsilon > 0$ such that for any $i \in \hat{n}$, $\exp\{\int_{t_0}^t y_i(s) ds\} \leq M \exp\{-\epsilon(t - t_0)\}$, $t \geq t_0$.

Proof: To prove this theorem, multiplying both sides of error neural networks (12) by $p|e_i(x, t)|^{p-2}e_i(x, t)$ and then integrating both sides with respect to the space variable x over the cube Ω yield

$$\begin{aligned} & \int_{\Omega} \frac{\partial |e_i(x, t)|^p}{\partial t} dx \\ & = \int_{\Omega} p|e_i(x, t)|^{p-2} e_i(x, t) \frac{\partial e_i(x, t)}{\partial t} dx \\ & \leq \int_{\Omega} p|e_i(x, t)|^{p-2} e_i(x, t) \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right] dx \\ & \quad - \int_{\Omega} p b_i |e_i(x, t)|^p dx \\ & \quad + \int_{\Omega} \sum_{j=1}^n \frac{r_j}{r_i} p |c_{ij}| l_j |e_i(x, t)|^{p-1} |e_j(x, t)| dx \\ & \quad + \int_{\Omega} \sum_{j=1}^n \frac{r_j}{r_i} p \delta_{ij} |e_i(x, t)|^{p-2} e_i(x, t) e_j(x, t) dx \\ & \quad + \int_{\Omega} \sum_{j=1}^n \frac{r_j}{r_i} p |h_{ij}| \tilde{l}_j |e_i(x, t)|^{p-1} |e_j(x, t - d_{ij}(t))| dx \\ & \leq \int_{\Omega} p|e_i(x, t)|^{p-2} e_i(x, t) \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right] dx \\ & \quad + \int_{\Omega} p(\delta_{ii} - b_i + |c_{ii}|l_i) |e_i(x, t)|^p dx \end{aligned}$$

$$\begin{aligned}
& + \sum_{j=1, j \neq i}^n \int_{\Omega} \frac{r_j}{r_i} p |c_{ij}| l_j |e_i(x, t)|^{p-1} |e_j(x, t)| dx \\
& + \sum_{j=1, j \neq i}^n \int_{\Omega} \frac{r_j}{r_i} p |\delta_{ij}| |e_i(x, t)|^{p-1} |e_j(x, t)| dx \\
& + \sum_{j=1}^n \int_{\Omega} \frac{r_j}{r_i} p |h_{ij}| \tilde{l}_j |e_i(x, t)|^{p-1} |e_j(x, t - d_{ij}(t))| dx \quad (24)
\end{aligned}$$

where condition (13) is utilized in the deducing process.

Applying the Green's formula [66, p. 628], to the reaction-diffusion term in (24) yields

$$\begin{aligned}
& \int_{\Omega} p |e_i(x, t)|^{p-2} e_i(x, t) \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right] dx \\
& = \int_{\Omega} p |e_i(x, t)|^{p-2} e_i(x, t) \nabla \cdot \left(a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right)_{k=1}^m dx \\
& = \int_{\Omega} \nabla \cdot \left(p |e_i(x, t)|^{p-2} e_i(x, t) a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right)_{k=1}^m dx \\
& \quad - p(p-1) \int_{\Omega} |e_i(x, t)|^{p-2} \sum_{k=1}^m a_{ik} \left[\frac{\partial e_i(x, t)}{\partial x_k} \right]^2 dx \\
& = \int_{\partial \Omega} \left(p |e_i(x, t)|^{p-2} e_i(x, t) a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right)_{k=1}^m dS \\
& \quad - p(p-1) \int_{\Omega} |e_i(x, t)|^{p-2} \sum_{k=1}^m a_{ik} \left[\frac{\partial e_i(x, t)}{\partial x_k} \right]^2 dx \quad (25)
\end{aligned}$$

in which $\nabla = [(\partial/\partial x_1), (\partial/\partial x_2), \dots, (\partial/\partial x_m)]^T$ represents the gradient operator, and

$$\left(a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right)_{k=1}^m = \left(a_{i1} \frac{\partial e_i(x, t)}{\partial x_1}, a_{i2} \frac{\partial e_i(x, t)}{\partial x_2}, \dots, a_{im} \frac{\partial e_i(x, t)}{\partial x_m} \right).$$

In view of boundary value condition (15) of error neural networks (12) and Lemma 2, one has

$$\begin{aligned}
& \int_{\Omega} p |e_i(x, t)|^{p-2} e_i(x, t) \sum_{k=1}^m \frac{\partial}{\partial x_k} \left[a_{ik} \frac{\partial e_i(x, t)}{\partial x_k} \right] dx \\
& = -p(p-1) \int_{\Omega} |e_i(x, t)|^{p-2} \sum_{k=1}^m a_{ik} \left[\frac{\partial e_i(x, t)}{\partial x_k} \right]^2 dx \\
& \leq - \sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} \int_{\Omega} |e_i(x, t)|^p dx. \quad (26)
\end{aligned}$$

On the other hand, as to $p |c_{ij}| l_j |e_i(x, t)|^{p-1} |e_j(x, t)|$ in (24), using Lemma 3 and condition (23) yield

$$\begin{aligned}
p |c_{ij}| l_j |e_i(x, t)|^{p-1} |e_j(x, t)| & = p \prod_{k=1}^{\Gamma_1} \left[|c_{ij}|^{\frac{\alpha_{ij}}{\mu_k}} l_j^{\frac{\beta_{ij}}{\mu_k}} |e_i(x, t)| \right]^{\mu_k} \\
& \quad \times \left[|c_{ij}|^{\alpha_{ij}^*} l_j^{\beta_{ij}^*} |e_j(x, t)| \right] \\
& \leq \sum_{k=1}^{\Gamma_1} \mu_k |c_{ij}|^{\frac{p\alpha_{ij}}{\mu_k}} l_j^{\frac{p\beta_{ij}}{\mu_k}} |e_i(x, t)|^p \\
& \quad + |c_{ij}|^{p\alpha_{ij}^*} l_j^{p\beta_{ij}^*} |e_j(x, t)|^p. \quad (27)
\end{aligned}$$

Similar to (27), utilizing condition (23), one immediately obtains

$$\begin{aligned}
& p |\delta_{ij}| |e_i(x, t)|^{p-1} |e_j(x, t)| \\
& = p \prod_{k=1}^{\Gamma_2} \left[|\delta_{ij}|^{\frac{\theta_{ij}}{\nu_k}} |e_i(x, t)| \right]^{\nu_k} \left[|\delta_{ij}|^{\theta_{ij}^*} |e_j(x, t)| \right] \\
& \leq \sum_{k=1}^{\Gamma_2} \nu_k |\delta_{ij}|^{\frac{p\theta_{ij}}{\nu_k}} |e_i(x, t)|^p + |\delta_{ij}|^{p\theta_{ij}^*} |e_j(x, t)|^p \quad (28)
\end{aligned}$$

and

$$\begin{aligned}
& p |h_{ij}| \tilde{l}_j |e_i(x, t)|^{p-1} |e_j(x, t - d_{ij}(t))| \\
& = p \prod_{k=1}^{\Gamma_3} \left[|h_{ij}|^{\frac{q_{ij}}{\omega_k}} \tilde{l}_j^{\frac{\sigma_{ij}}{\omega_k}} |e_i(x, t)| \right]^{\omega_k} \\
& \quad \times \left[|h_{ij}|^{q_{ij}^*} \tilde{l}_j^{\sigma_{ij}^*} |e_j(x, t - d_{ij}(t))| \right] \\
& \leq \sum_{k=1}^{\Gamma_3} \omega_k |h_{ij}|^{\frac{pq_{ij}}{\omega_k}} \tilde{l}_j^{\frac{p\sigma_{ij}}{\omega_k}} |e_i(x, t)|^p \\
& \quad + |h_{ij}|^{pq_{ij}^*} \tilde{l}_j^{p\sigma_{ij}^*} |e_j(x, t - d_{ij}(t))|^p. \quad (29)
\end{aligned}$$

Substituting (26)–(29) into (24) gives

$$\begin{aligned}
& \int_{\Omega} \frac{\partial |e_i(x, t)|^p}{\partial t} dx \\
& \leq \int_{\Omega} \left[- \sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} + p(\delta_{ii} - b_i + |c_{ii}| l_i) \right. \\
& \quad + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_1} \frac{r_j}{r_i} \mu_k |c_{ij}|^{\frac{p\alpha_{ij}}{\mu_k}} l_j^{\frac{p\beta_{ij}}{\mu_k}} \\
& \quad + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_2} \frac{r_j}{r_i} \nu_k |\delta_{ij}|^{\frac{p\theta_{ij}}{\nu_k}} \\
& \quad + \sum_{j=1}^n \sum_{k=1}^{\Gamma_3} \frac{r_j}{r_i} \omega_k |h_{ij}|^{\frac{pq_{ij}}{\omega_k}} \tilde{l}_j^{\frac{p\sigma_{ij}}{\omega_k}} \left. \right] |e_i(x, t)|^p dx \\
& \quad + \sum_{j=1, j \neq i}^n \int_{\Omega} \frac{r_j}{r_i} \left[|c_{ij}|^{p\alpha_{ij}^*} l_j^{p\beta_{ij}^*} + |\delta_{ij}|^{p\theta_{ij}^*} \right] |e_j(x, t)|^p dx \\
& \quad + \sum_{j=1}^n \int_{\Omega} \frac{r_j}{r_i} |h_{ij}|^{pq_{ij}^*} \tilde{l}_j^{p\sigma_{ij}^*} |e_j(x, t - d_{ij}(t))|^p dx. \quad (30)
\end{aligned}$$

Now define $\xi_i(t) = \int_{\Omega} |e_i(x, t)|^p dx$ and $\xi(t_0) = \max_{1 \leq i \leq n} \sup_{-\infty < s \leq t_0} \xi_i(s)$. Then, choose adequate functions $z_i(t) \in \mathbb{C}^0(\mathbb{R}, \mathbb{R})$ such that $y_i(s) + z_i(s) \leq 0$ and $z_i(s) \geq 0$ for all $s \leq t_0$, $i \in \hat{n}$. Additionally, define functions

$$\pi_i(t) = \begin{cases} \xi_i(t) \exp \left\{ \int_{t_0}^t z_i(s) ds \right\}, & t \geq t_0 \\ \xi_i(t), & t < t_0 \end{cases} \quad (31)$$

$i \in \hat{n}$, and

$$\chi_i(t) = \pi_i(t) - \xi(t_0) \exp \left\{ \int_{t_0}^t [y_i(s) + z_i(s)] ds \right\} \quad (32)$$

$t \in \mathbb{R}$, $i \in \hat{n}$. It is clear that the function $\pi_i(t)$ is continuous for $t \in \mathbb{R} \setminus \{t_0\}$. Meanwhile, $\pi_i(t)$ is also continuous when $t = t_0$ since $\lim_{t \rightarrow t_0^+} \pi_i(t) = \lim_{t \rightarrow t_0^-} \pi_i(t) = \xi_i(t_0) = \int_{\Omega} |e_i(x, t_0)|^p dx$, $i \in \hat{n}$. Hence, $\chi_i(t)$ is also a continuous function. It should be noted that for any $i \in \hat{n}$, $\chi_i(s) \leq 0$ when $s < t_0$ due to the properties of $y_i(s) + z_i(s)$. More importantly, $\chi_i(t) \leq 0$ also holds for $t \geq t_0$, $i \in \hat{n}$. If this claim is not true, then there exist $t_1 \geq t_0$ and $\iota \in \hat{n}$ such that

$$\chi_{\iota}(t_1) = 0, \quad \mathbb{D}^+ \chi_{\iota}(t)|_{t=t_1} \geq 0 \quad (33)$$

and for all $j \in \hat{n}$,

$$\chi_j(s) \leq 0, \quad s < t_1. \quad (34)$$

Now we investigate the term $\mathbb{D}^+ \chi_{\iota}(t)|_{t=t_1}$, it follows from (30)–(32) that

$$\begin{aligned} \mathbb{D}^+ \chi_{\iota}(t)|_{t=t_1} &= \mathbb{D}^+ \left\{ \xi_{\iota}(t) \exp \left[\int_{t_0}^t z_{\iota}(s) ds \right] \right. \\ &\quad \left. - \xi(t_0) \exp \left[\int_{t_0}^t y_{\iota}(s) + z_{\iota}(s) ds \right] \right\} \Big|_{t=t_1} \\ &\leq \left\{ \left[- \sum_{k=1}^m \frac{4(p-1)a_{\iota k}}{pq_k^2} + p(\delta_{\iota\iota} - b_{\iota} + |c_{\iota\iota}|l_{\iota}) \right. \right. \\ &\quad + \sum_{j=1, j \neq \iota}^n \sum_{k=1}^{\Gamma_1} \frac{r_j}{r_{\iota}} \mu_k |c_{ij}|^{\frac{p\alpha_{ij}}{\mu_k}} l_j^{\frac{p\beta_{ij}}{\mu_k}} \\ &\quad + \sum_{j=1, j \neq \iota}^n \sum_{k=1}^{\Gamma_2} \frac{r_j}{r_{\iota}} \nu_k |\delta_{ij}|^{\frac{p\theta_{ij}}{\nu_k}} \\ &\quad \left. + \sum_{j=1}^n \sum_{k=1}^{\Gamma_3} \frac{r_j}{r_{\iota}} \omega_k |h_{ij}|^{\frac{p\varrho_{ij}}{\omega_k}} \tilde{l}_j^{\frac{p\sigma_{ij}}{\omega_k}} \right] \xi_{\iota}(t_1) \\ &\quad + \sum_{j=1, j \neq \iota}^n \left\{ \frac{r_j}{r_{\iota}} \left[|c_{ij}|^{p\alpha_{ij}^*} l_j^{p\beta_{ij}^*} + |\delta_{ij}|^{p\theta_{ij}^*} \right] \xi_j(t_1) \right\} \\ &\quad + \sum_{j=1}^n \frac{r_j}{r_{\iota}} |h_{ij}|^{p\varrho_{ij}^*} \tilde{l}_j^{p\sigma_{ij}^*} \xi_j(t_1 - d_{ij}(t_1)) \\ &\quad + \xi_{\iota}(t_1) z_{\iota}(t_1) - \xi(t_0) [y_{\iota}(t_1) + z_{\iota}(t_1)] \\ &\quad \times \exp \left[\int_{t_0}^{t_1} y_{\iota}(s) ds \right] \Big\} \exp \left\{ \int_{t_0}^{t_1} z_{\iota}(s) ds \right\}. \quad (35) \end{aligned}$$

Recall that when $t_1 - d_{ij}(t_1) \geq t_0$, one obtains $\xi_j(t_1 - d_{ij}(t_1)) \leq \xi(t_0) \exp \left\{ \int_{t_0}^{t_1 - d_{ij}(t_1)} y_j(s) ds \right\}$ on the basis of (33) and (34). In fact, $t_1 - d_{ij}(t_1) < t_0$ may hold. In this case, one has $\xi_j(t_1 - d_{ij}(t_1)) \leq \xi(t_0) \exp \left\{ \int_{t_0}^{t_1 - d_{ij}(t_1)} [y_j(s) + z_j(s)] ds \right\}$. Note

that $z_j(s) \geq 0$ when $s \leq t_0$. Hence, one gets

$$\begin{aligned} \mathbb{D}^+ \chi_{\iota}(t)|_{t=t_1} &\leq \left\{ \left[- \sum_{k=1}^m \frac{4(p-1)a_{\iota k}}{pq_k^2} + p(\delta_{\iota\iota} - b_{\iota} + |c_{\iota\iota}|l_{\iota}) \right. \right. \\ &\quad + \sum_{j=1, j \neq \iota}^n \sum_{k=1}^{\Gamma_1} \frac{r_j}{r_{\iota}} \mu_k |c_{ij}|^{\frac{p\alpha_{ij}}{\mu_k}} l_j^{\frac{p\beta_{ij}}{\mu_k}} \\ &\quad + \sum_{j=1, j \neq \iota}^n \sum_{k=1}^{\Gamma_2} \frac{r_j}{r_{\iota}} \nu_k |\delta_{ij}|^{\frac{p\theta_{ij}}{\nu_k}} \\ &\quad \left. + \sum_{j=1}^n \sum_{k=1}^{\Gamma_3} \frac{r_j}{r_{\iota}} \omega_k |h_{ij}|^{\frac{p\varrho_{ij}}{\omega_k}} \tilde{l}_j^{\frac{p\sigma_{ij}}{\omega_k}} \right] \exp \left\{ \int_{t_0}^{t_1} y_{\iota}(s) ds \right\} \\ &\quad + \sum_{j=1, j \neq \iota}^n \left\{ \frac{r_j}{r_{\iota}} \left[|c_{ij}|^{p\alpha_{ij}^*} l_j^{p\beta_{ij}^*} + |\delta_{ij}|^{p\theta_{ij}^*} \right] \right. \\ &\quad \times \exp \left\{ \int_{t_0}^{t_1} y_j(s) ds \right\} \Big\} \\ &\quad + \sum_{j=1}^n \frac{r_j}{r_{\iota}} |h_{ij}|^{p\varrho_{ij}^*} \tilde{l}_j^{p\sigma_{ij}^*} \exp \left\{ \int_{t_0}^{t_1 - d_{ij}(t_1)} y_j(s) ds \right\} \\ &\quad - y_{\iota}(t_1) \exp \left\{ \int_{t_0}^{t_1} y_{\iota}(s) ds \right\} \Big\} \\ &\quad \times \xi(t_0) \exp \left\{ \int_{t_0}^{t_1} z_{\iota}(s) ds \right\}. \quad (36) \end{aligned}$$

In light of conditions (22) and (23), one has $\mathbb{D}^+ \chi_{\iota}(t)|_{t=t_1} < 0$, which contradicts with condition (33). Thus, for any $t \geq t_0$, $\chi_i(t) \leq 0$ holds for every $i \in \hat{n}$. Combining with the definition in (32), therefore, one gets

$$\|e_i(x, t)\|^p \leq \xi(t_0) \exp \left\{ \int_{t_0}^t y_i(s) ds \right\} \quad (37)$$

$t \geq t_0$, $i \in \hat{n}$. If $\lim_{t \rightarrow +\infty} \int_{t_0}^t y_i(s) ds = -\infty$, then $\lim_{t \rightarrow +\infty} \|e_i(x, t)\|^p = 0$, $i \in \hat{n}$, which implies that neural networks (12) are globally asymptotically stable. Furthermore, if $\exp \left\{ \int_{t_0}^t y_i(s) ds \right\} \leq M \exp \{-\epsilon(t - t_0)\}$, it follows from (37) that $\|e_i(x, t)\|^p \leq \max_{1 \leq i \leq n} \sup_{-\infty < s \leq t_0} \|e_i(x, s)\|^p M \exp \{-\epsilon(t - t_0)\}$, $t \geq t_0$, $i \in \hat{n}$. Hence, neural networks (12) are globally exponentially stable. The proof is now completed. ■

Remark 4: To investigate synchronization of various neural networks, the Lyapunov's method is utilized in [20], [28], [30], [37], [39], [44], [46], [49], and [54]–[62]. To do this, a Lyapunov function or functional should be constructed, then we calculate time derivative of the chosen function or functional along the trajectories of the considered neural networks. In reality, it is a difficult task when we try to build an appropriate Lyapunov function or functional. Meanwhile, how to tightly estimate derivative terms is hard. In this paper, Theorem 1 provides an approach to investigate synchronization for master neural networks (2) and slave neural networks (7) under the designed controller (8) on the basis of the concept of comparison. When (22) and (23) hold, stability of neural networks (12) can be ascertained.

Remark 5: In delayed reaction-diffusion neural networks (2), (7), and (12), time-varying delays $d_{ij}(t)$, $i, j \in \hat{n}$ are assumed to be infinite. In this paper, Theorem 1 presents techniques to study stability of general neural networks with infinite delays.

To deal with the cross terms in (27)–(29), Lemma 3 is utilized. Now a corollary is provided by using Lemma 1.

Corollary 1: Given constants $p \geq 2$, q_s , $s \in \hat{m}$ and l_i , $\tilde{l}_i$, $i \in \hat{n}$. Suppose that Assumptions 1 and 2 are satisfied. Neural networks (12) are globally asymptotically stable, namely, master neural networks (2) and slave neural networks (7) are globally asymptotically synchronized under the designed controller (8) in the sense of Definition 1, if there exist functions $\tilde{y}_i(t) \in \mathbb{C}^0(\mathbb{R}, \mathbb{R})$ with $\tilde{y}_i(s) \leq 0$, $\forall s \leq t_0$, and $\lim_{t \rightarrow +\infty} \int_{t_0}^t \tilde{y}_i(s) ds = -\infty$, $i \in \hat{n}$, positive constants $\tilde{r}_k$, $k \in \hat{n}$, negative constants $\tilde{\delta}_{ii}$, $i \in \hat{n}$ and any real constants $\tilde{\delta}_{ij}$, $i \neq j$, $i, j \in \hat{n}$ such that for any $t \geq t_0 > 0$

$$\begin{aligned} & \left[-\sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} + p(\tilde{\delta}_{ii} - b_i + |c_{ii}|l_i) \right. \\ & + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} (p-1) \left[|c_{ij}|l_j + |\tilde{\delta}_{ij}| \right] \\ & + \sum_{j=1}^n \frac{\tilde{r}_j}{\tilde{r}_i} (p-1) |h_{ij}| \tilde{l}_j - \tilde{y}_i(t) \left. \right] \exp \left\{ \int_{t_0}^t \tilde{y}_i(s) ds \right\} \\ & + \sum_{j=1, j \neq i}^n \left\{ \frac{\tilde{r}_j}{\tilde{r}_i} \left[|c_{ij}|l_j + |\tilde{\delta}_{ij}| \right] \exp \left\{ \int_{t_0}^t \tilde{y}_j(s) ds \right\} \right\} \\ & + \sum_{j=1}^n \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}| \tilde{l}_j \exp \left\{ \int_{t_0}^{t-d_{ij}(t)} \tilde{y}_j(s) ds \right\} < 0. \quad (38) \end{aligned}$$

Furthermore, neural networks (12) are globally exponentially stable, that is, neural networks (2) and (7) are globally exponentially synchronized under the designed controller (8) in the sense of Definition 1, if there exist $\tilde{M} > 0$ and $\tilde{\epsilon} > 0$ such that for any $i \in \hat{n}$, $\exp \left\{ \int_{t_0}^t \tilde{y}_i(s) ds \right\} \leq \tilde{M} \exp \{-\tilde{\epsilon}(t - t_0)\}$, $t \geq t_0$.

Proof: It follows from Lemma 1 that:

$$\begin{aligned} & p|c_{ij}|l_j|e_i(x, t)|^{p-1}|e_j(x, t)| \\ & \leq p|c_{ij}|l_j \left[\frac{1}{q}|e_i(x, t)|^{(p-1)q} + \frac{1}{p}|e_j(x, t)|^p \right] \\ & = |c_{ij}|l_j \left[(p-1)|e_i(x, t)|^p + |e_j(x, t)|^p \right]. \quad (39) \end{aligned}$$

Similarly, one has

$$\begin{aligned} & p|\tilde{\delta}_{ij}||e_i(x, t)|^{p-1}|e_j(x, t)| \\ & \leq p|\tilde{\delta}_{ij}| \left[\frac{1}{q}|e_i(x, t)|^{(p-1)q} + \frac{1}{p}|e_j(x, t)|^p \right] \\ & = |\tilde{\delta}_{ij}| \left[(p-1)|e_i(x, t)|^p + |e_j(x, t)|^p \right] \quad (40) \end{aligned}$$

and

$$\begin{aligned} & p|h_{ij}|\tilde{l}_j|e_i(x, t)|^{p-1}|e_j(x, t-d_{ij}(t))| \\ & \leq p|h_{ij}|\tilde{l}_j \left[\frac{1}{q}|e_i(x, t)|^{(p-1)q} + \frac{1}{p}|e_j(x, t-d_{ij}(t))|^p \right] \\ & = |h_{ij}|\tilde{l}_j \left[(p-1)|e_i(x, t)|^p + |e_j(x, t-d_{ij}(t))|^p \right]. \quad (41) \end{aligned}$$

Then, similar to the proof in Theorem 1, one can get

$$\begin{aligned} & \int_{\Omega} \frac{\partial |e_i(x, t)|^p}{\partial t} dx \\ & \leq \int_{\Omega} \left[-\sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} + p(\tilde{\delta}_{ii} - b_i + |c_{ii}|l_i) \right. \\ & + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} (p-1) |c_{ij}|l_j \\ & + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} (p-1) |\tilde{\delta}_{ij}| \\ & + \sum_{j=1}^n \frac{\tilde{r}_j}{\tilde{r}_i} (p-1) |h_{ij}|\tilde{l}_j \left. \right] |e_i(x, t)|^p dx \\ & + \sum_{j=1, j \neq i}^n \int_{\Omega} \frac{\tilde{r}_j}{\tilde{r}_i} \left[|c_{ij}|l_j + |\tilde{\delta}_{ij}| \right] |e_j(x, t)|^p dx \\ & + \sum_{j=1}^n \int_{\Omega} \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}|\tilde{l}_j |e_j(x, t-d_{ij}(t))|^p dx. \quad (42) \end{aligned}$$

The following proof is similar to that of Theorem 1, so the detailed process is omitted here. ■

In the following part of this section, time-varying delays $d_{ij}(t)$, $i, j \in \hat{n}$ are assumed to be bounded [i.e., there exists a constant $d > 0$ such that $d_{ij}(t) \leq d$]. In light of Theorem 1, novel criteria are provided to ensure exponential synchronization of master neural networks (2) and slave neural networks (7) under the designed controller (8).

Corollary 2: Given constants $p \geq 2$, q_s , $s \in \hat{m}$ and l_i , $\tilde{l}_i$, $i \in \hat{n}$. Suppose that Assumption 1 is satisfied and let $d_{ij}(t)$, $i, j \in \hat{n}$ be bounded functions. Neural networks (12) are globally exponentially stable, namely, master neural networks (2) and slave neural networks (7) are globally exponentially synchronized under the designed controller (8) in the sense of Definition 1, if there exist positive integers Γ_4 , Γ_5 , Γ_6 , positive constants $\check{\mu}_k$, $k \in \hat{\Gamma}_4$, $\check{\nu}_k$, $k \in \hat{\Gamma}_5$, $\check{\omega}_k$, $k \in \hat{\Gamma}_6$, $\check{r}_k$, $k \in \hat{n}$, ϑ , negative constants $\check{\delta}_{ii}$, $i \in \hat{n}$ and any real constants $\check{\delta}_{ij}$, $i \neq j$, $\check{\alpha}_{ij}$, $\check{\beta}_{ij}^*$, $\check{\beta}_{ij}^*$, $\check{\theta}_{ij}$, $\check{\theta}_{ij}^*$, $\check{\varrho}_{ij}$, $\check{\varrho}_{ij}^*$, $\check{\sigma}_{ij}$, $\check{\sigma}_{ij}^*$, $i, j \in \hat{n}$ such that for any $t \geq t_0 > 0$

$$\begin{aligned} & \left[\vartheta - \sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} + p(\check{\delta}_{ii} - b_i + |c_{ii}|l_i) \right. \\ & + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_4} \frac{\check{r}_j}{\check{r}_i} \check{\mu}_k |c_{ij}|^{\frac{p\check{\alpha}_{ij}}{\check{\mu}_k}} l_j^{\frac{p\check{\beta}_{ij}}{\check{\mu}_k}} + \sum_{j=1, j \neq i}^n \sum_{k=1}^{\Gamma_5} \frac{\check{r}_j}{\check{r}_i} \check{\nu}_k |\check{\delta}_{ij}|^{\frac{p\check{\theta}_{ij}}{\check{\nu}_k}} \\ & + \sum_{j=1}^n \sum_{k=1}^{\Gamma_6} \frac{\check{r}_j}{\check{r}_i} \check{\omega}_k |h_{ij}|^{\frac{p\check{\varrho}_{ij}}{\check{\omega}_k}} \tilde{l}_j^{\frac{p\check{\sigma}_{ij}}{\check{\omega}_k}} \left. \right] \\ & + \sum_{j=1, j \neq i}^n \frac{\check{r}_j}{\check{r}_i} \left[|c_{ij}|^{\frac{p\check{\alpha}_{ij}}{\check{r}_i}} l_j^{\frac{p\check{\beta}_{ij}}{\check{r}_i}} + |\check{\delta}_{ij}|^{\frac{p\check{\theta}_{ij}}{\check{r}_i}} \right] \\ & + \sum_{j=1}^n \frac{\check{r}_j}{\check{r}_i} |h_{ij}|^{\frac{p\check{\varrho}_{ij}}{\check{r}_i}} \tilde{l}_j^{\frac{p\check{\sigma}_{ij}}{\check{r}_i}} \exp \{ \vartheta d_{ij}(t) \} < 0 \quad (43) \end{aligned}$$

in which

$$\begin{aligned}\Gamma_4 \check{\alpha}_{ij} + \check{\alpha}_{ij}^* &= 1, \quad \Gamma_4 \check{\beta}_{ij} + \check{\beta}_{ij}^* = 1, \quad \Gamma_5 \check{\theta}_{ij} + \check{\theta}_{ij}^* = 1 \\ \Gamma_6 \check{\varrho}_{ij} + \check{\varrho}_{ij}^* &= 1, \quad \Gamma_6 \check{\sigma}_{ij} + \check{\sigma}_{ij}^* = 1 \\ \sum_{k=1}^{\Gamma_4} \check{\mu}_k &= \sum_{k=1}^{\Gamma_5} \check{\nu}_k = \sum_{k=1}^{\Gamma_6} \check{\omega}_k = p - 1.\end{aligned}\quad (44)$$

Proof: Let $y_i(t) = -\vartheta$, $i \in \hat{n}$, according to (22) and (43), exponential stability of error neural networks (12) can be collected. Additionally, exponential synchronization of master neural networks (2) and slave neural networks (7) can be realized. So the detailed proof is omitted here. ■

Now we turn to investigate global exponential stability of neural networks (2). Based on [52], an M -matrix-based criterion is given.

Corollary 3: Given constants $p \geq 2$, q_s , $s \in \hat{m}$ and l_i , $\tilde{l}_i$, $i \in \hat{n}$. Suppose that Assumption 1 is satisfied and let $d_{ij}(t)$, $i, j \in \hat{n}$ be bounded functions. Neural networks (2) are globally exponentially stable, if matrix $\mathbb{S} = [s_{ij}]_{n \times n}$ is an M -matrix with $s_{ii} = \sum_{k=1}^m (4(p-1)a_{ik}/pq_k^2) + p(b_i - |c_{ii}l_i - |h_{ii}\tilde{l}_i|)$, and $s_{ij} = -p|c_{ij}l_j - p|h_{ij}\tilde{l}_j|$, $i \neq j$, $i, j \in \hat{n}$.

Proof: Since the matrix $\mathbb{S}$ is an M -matrix, based on Lemma 4, there exist constants $\tilde{r}_i > 0$, $i \in \hat{n}$ such that

$$\begin{aligned}-\sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} - pb_i + p|c_{ii}l_i + p|h_{ii}\tilde{l}_i \\ + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} p|c_{ij}l_j + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} p|h_{ij}\tilde{l}_j < 0.\end{aligned}\quad (45)$$

Note that $d_{ij}(t)$ is bounded, $i, j \in \hat{n}$, there surely exists a sufficiently small constant $\epsilon > 0$ such that

$$\begin{aligned}\epsilon - \sum_{k=1}^m \frac{4(p-1)a_{ik}}{pq_k^2} - pb_i + p|c_{ii}l_i \\ + \sum_{j=1, j \neq i}^n (p-1) \frac{\tilde{r}_j}{\tilde{r}_i} |c_{ij}l_j + \sum_{j=1}^n (p-1) \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}\tilde{l}_j \\ + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} |c_{ij}l_j + \sum_{j=1}^n \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}\tilde{l}_j \exp\{\epsilon d_{ij}(t)\} < 0.\end{aligned}\quad (46)$$

Combining with Corollaries 1 and 2, the conclusion is clear. So the mathematical proof is omitted here. ■

Remark 6: Liang and Cao [52] indicate that under the condition that $B - (|C| + |H|)L$ is an M -matrix, in which $B = \text{diag}\{b_1, b_2, \dots, b_n\}$, $|C| = [|c_{ij}|]_{n \times n}$, $|H| = [|h_{ij}|]_{n \times n}$ and $L = \text{diag}\{l_1, l_2, \dots, l_n\}$, neural networks (2) are globally exponentially stable. Recall that when $p = 2$, $l_i = \tilde{l}_i$, $i \in \hat{n}$, from Corollary 3

$$\begin{aligned}-\sum_{k=1}^m \frac{a_{ik}}{q_k^2} - b_i + |c_{ii}l_i + |h_{ii}\tilde{l}_i| \\ + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} |c_{ij}l_j + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}\tilde{l}_j \leq -b_i + |c_{ii}l_i + |h_{ii}\tilde{l}_i| \\ + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} |c_{ij}l_j + \sum_{j=1, j \neq i}^n \frac{\tilde{r}_j}{\tilde{r}_i} |h_{ij}\tilde{l}_j < 0, \quad i \in \hat{n}.\end{aligned}\quad (47)$$

Hence, the criterion in [52] is more restrictive than the one in Corollary 3. Meanwhile, in [4], [5], and [15], the scholars show that if $B - |C|L - |H|\tilde{L}$ is an M -matrix, then neural networks (5) are globally exponentially stable. Let $a_{ik} = 0$, $i \in \hat{n}$, $k \in \hat{m}$ and $p = 2$, then Corollary 3 reduces to them.

Remark 7: In Corollary 3, an M -matrix-based criterion is developed to analyze stability of neural networks (2). Recall that linear matrix inequality (LMI) technique is also effective to investigate stability of neural networks, please refer to [36]. Yet, the maximal order of the LMIs and the number of decision variables influence calculation complexity when using the LMI toolbox in MATLAB. By contrast, M -matrix-based criteria can be verified easily without utilizing the MATLAB toolbox.

IV. NUMERICAL EXAMPLES

In this section, three numerical examples are performed to illustrate the effectiveness and merits of the proposed theoretical results. The first one is to the case of infinite delays, while the last two are to the ones of finite delays.

Example 1: Consider master neural networks (2) and slave neural networks (7) with following parameters: $m = 1$, $i = j = 2$, $a_1 = a_2 = 1$, $b_1 = b_2 = 2$

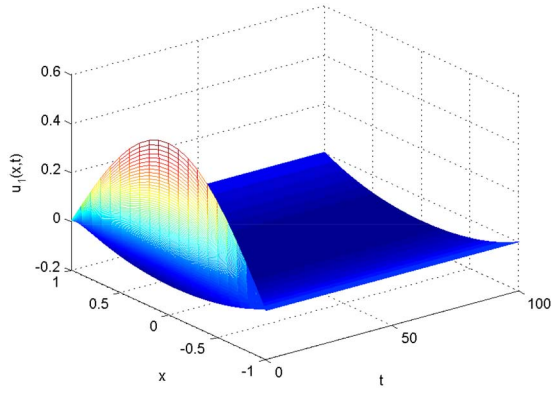
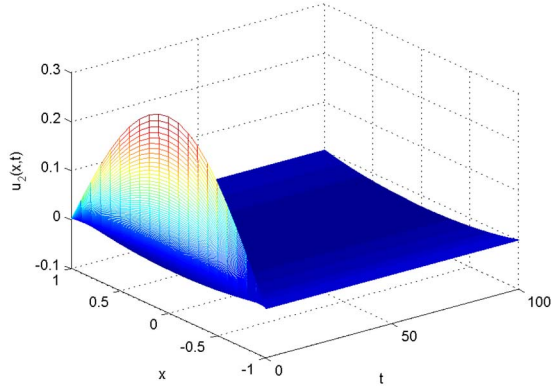
$$C = [c_{ij}]_{2 \times 2} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{3} \\ \frac{1}{4} & -\frac{1}{2} \end{bmatrix}$$

$$H = [h_{ij}]_{2 \times 2} = \begin{bmatrix} \frac{1}{2} & \frac{1}{5} \\ -\frac{1}{3} & \frac{1}{2} \end{bmatrix}$$

infinite time-varying delays $d_{11}(t) = (3t/4)$, $d_{12}(t) = (9t/10)$, $d_{21}(t) = (5t/6)$, $d_{22}(t) = (3t/4)$, $p = 2$, activation functions $\tilde{f}_1(u) = \tilde{f}_2(u) = \tilde{g}_1(u) = \tilde{g}_2(u) = (1/4)(|u+1| - |u-1|)$, $x \in \Omega = [-1, 1]$, and $t \geq t_0 = 1$. Figs. 1 and 2 are state trajectories of $u_1(x, t)$ and $u_2(x, t)$, respectively. According to the numerical simulations in Figs. 1 and 2, it is clear that $u_1(x, t)$ and $u_2(x, t)$ are not convergent as time passes.

As to neural networks (12), based on the chosen activation functions $\tilde{f}_i(u)$ and $\tilde{g}_i(u)$, we know $l_i = \tilde{l}_i = (1/2)$, $i = 1, 2$. On the other hand, since $\Omega = [-1, 1]$, we have $q_1 = 1$. Then, choosing $\Gamma_1 = \Gamma_2 = \Gamma_3 = 1$, $\mu_1 = \nu_1 = \omega_1 = 1$, $r_1 = r_2 = 1$, $\alpha_{ij} = \alpha_{ij}^* = \beta_{ij} = \beta_{ij}^* = \theta_{ij} = \theta_{ij}^* = \varrho_{ij} = \varrho_{ij}^* = \sigma_{ij} = \sigma_{ij}^* = (1/2)$, $i, j = 1, 2$, $\delta_{11} = -1$, $\delta_{12} = (5/6)$, $\delta_{21} = (7/8)$, $\delta_{22} = -1$, and $y_i(t) = -(1/t)$ when $t \geq 1$, otherwise, it is -1 for $i = 1, 2$, combining with the proposed criteria in Theorem 1, one obtains the following results:

$$\begin{aligned}\left[-2 - \frac{11}{2} + \frac{1}{6} + \frac{5}{6} + \frac{1}{4} + \frac{1}{10} - \frac{1}{t}\right] \exp\left\{\int_{t_0}^t \frac{-1}{s} ds\right\} \\ + \exp\left\{\int_{t_0}^t \frac{-1}{s} ds\right\} + \frac{1}{4} \times \exp\left\{\int_{t_0}^{t-\frac{3t}{4}} \frac{-1}{s} ds\right\} \\ + \frac{1}{10} \times \exp\left\{\int_{t_0}^{t-\frac{9t}{10}} \frac{-1}{s} ds\right\} < 0\end{aligned}\quad (48)$$

Fig. 1. State trajectories of $u_1(x, t)$ in Example 1.Fig. 2. State trajectories of $u_2(x, t)$ in Example 1.

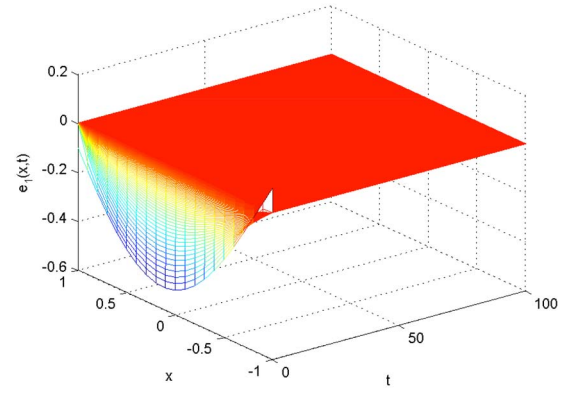
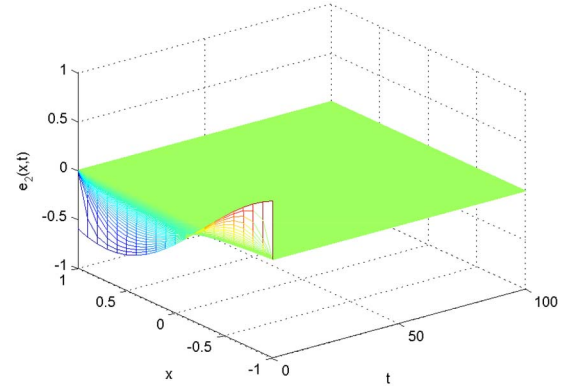
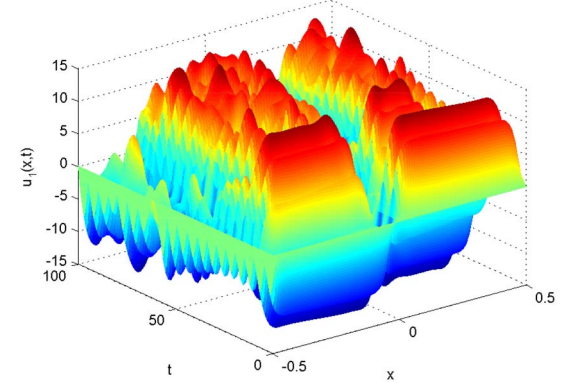
$$\begin{aligned} & \left[-2 - \frac{11}{2} + \frac{1}{8} + \frac{7}{8} + \frac{1}{6} + \frac{1}{4} - \frac{1}{t} \right] \exp \left\{ \int_{t_0}^t \frac{-1}{s} ds \right\} \\ & + \exp \left\{ \int_{t_0}^t \frac{-1}{s} ds \right\} + \frac{1}{6} \times \exp \left\{ \int_{t_0}^{t-\frac{5t}{6}} \frac{-1}{s} ds \right\} \\ & + \frac{1}{4} \times \exp \left\{ \int_{t_0}^{t-\frac{3t}{4}} \frac{-1}{s} ds \right\} < 0. \end{aligned} \quad (49)$$

Hence, from Theorem 1, we can conclude that error neural networks (12) are globally asymptotically stable. Therefore, master neural networks (2) and slave neural networks (7) are globally asymptotically synchronized under the designed controller (8). Figs. 3 and 4 are state trajectories of synchronization errors $e_1(x, t)$ and $e_2(x, t)$, respectively. Based on the numerical simulations in Figs. 3 and 4, it is clear that master neural networks (2) can realize synchronization with slave neural networks (7) under the designed controller (8). Since the time delays are unbounded, the proposed criteria in [33], [34], [36], [49], [52], and [53] are invalid.

Different from Example 1, in the following example, chaotic neural networks with finite delays are provided, which are borrowed from [49].

Example 2: Consider master neural networks (2) and slave neural networks (7) with following parameters: $m = 1$, $i = j = 3$, $a_1 = a_2 = a_3 = 10^{-4}$, $b_1 = b_2 = b_3 = 1$

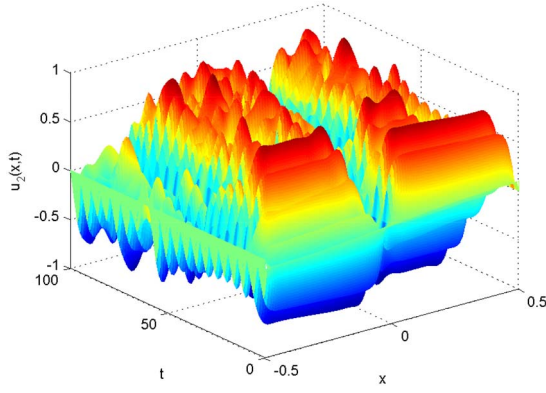
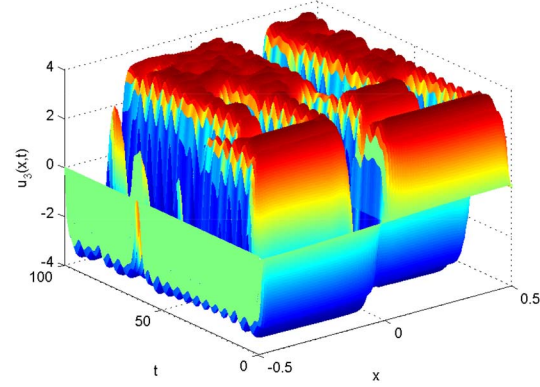
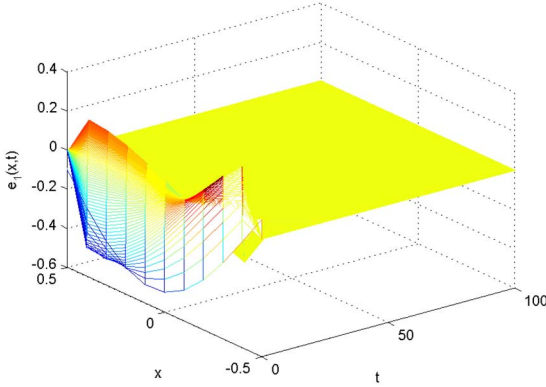
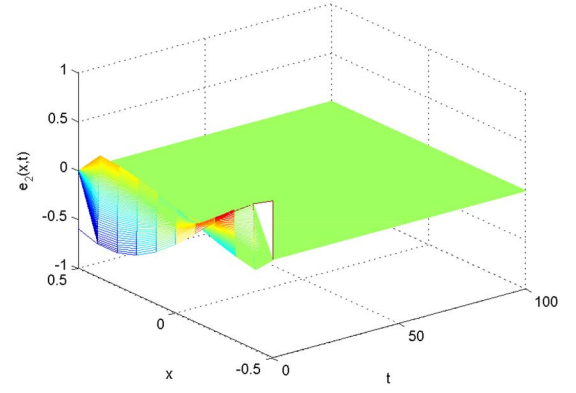
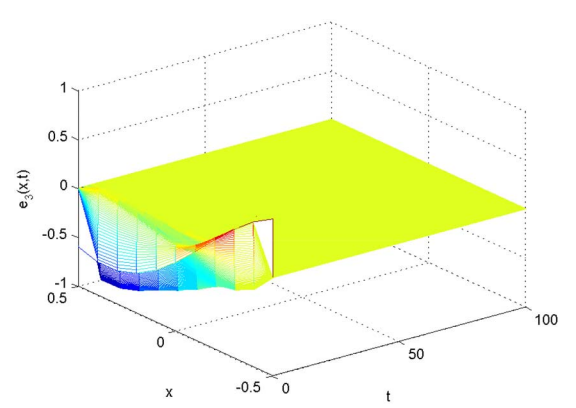
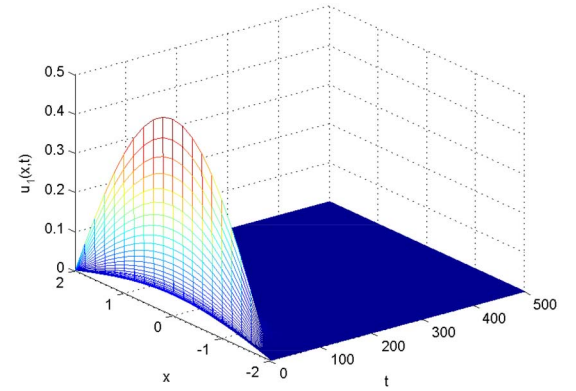
$$C = [c_{ij}]_{3 \times 3} = \begin{bmatrix} 1 + \frac{\pi}{4} & 20 & 0.001 \\ 0.1 & 1 + \frac{\pi}{4} & 0.001 \\ 3 & -0.56 & -0.12 \end{bmatrix}$$

Fig. 3. Synchronization error $e_1(x, t)$ in Example 1.Fig. 4. Synchronization error $e_2(x, t)$ in Example 1.Fig. 5. State trajectories of $u_1(x, t)$ in Example 2.

$$H = [h_{ij}]_{3 \times 3} = \begin{bmatrix} -1.3\sqrt{2}\frac{\pi}{4} & 0.1 & -0.001 \\ 0.1 & -1.3\sqrt{2}\frac{\pi}{4} & 0.01 \\ 2 & -0.85 & 0.02 \end{bmatrix}$$

$\tilde{f}_i(u) = \tilde{g}_i(u) = (1/2)(|u + 1| - |u - 1|)$, $i = 1, 2, 3$, $p = 2$, time delays $d_{ij}(t) = 0.9$, $i, j = 1, 2, 3$, $x \in \Omega = [-0.5, 0.5]$. Figs. 5–7 are state trajectories of $u_1(x, t)$, $u_2(x, t)$ and $u_3(x, t)$, respectively. As shown in Figs. 5–7, master neural networks (2) exhibit chaotic behaviors which can be utilized to secure communications.

On the other hand, based on the chosen functions $\tilde{f}_i(u)$ and $\tilde{g}_i(u)$, we know $l_i = \tilde{l}_i = 1$, $i = 1, 2, 3$. At the same time, according to $\Omega = [-0.5, 0.5]$, we have $q_1 = 0.5$.

Fig. 6. State trajectories of $u_2(x, t)$ in Example 2.Fig. 7. State trajectories of $u_3(x, t)$ in Example 2.Fig. 8. Synchronization error $e_1(x, t)$ in Example 2.Fig. 9. Synchronization error $e_2(x, t)$ in Example 2.Fig. 10. Synchronization error $e_3(x, t)$ in Example 2.Fig. 11. State trajectories of $u_1(x, t)$ in Example 3.

Then, choosing $\Gamma_4 = \Gamma_5 = \Gamma_6 = 1$, $\check{\mu}_1 = \check{\nu}_1 = \check{\omega}_1 = 1$, $\check{r}_1 = \check{r}_2 = \check{r}_3 = 1$, $\check{\alpha}_{ij} = \check{\alpha}_{ij}^* = \check{\beta}_{ij} = \check{\beta}_{ij}^* = \check{\theta}_{ij} = \check{\theta}_{ij}^* = \check{\varrho}_{ij} = \check{\varrho}_{ij}^* = \check{\sigma}_{ij} = \check{\sigma}_{ij}^* = (1/2)$, $i = 1, 2, 3$, $j = 1, 2, 3$, $\vartheta = 1$, $\check{\delta}_{11} = -25$, $\check{\delta}_{12} = -0.1$, $\check{\delta}_{13} = 0.2$, $\check{\delta}_{21} = 0.01$, $\check{\delta}_{22} = -5$, $\check{\delta}_{23} = 0.1$, $\check{\delta}_{31} = 0.2$, $\check{\delta}_{32} = 0.2$, and $\check{\delta}_{33} = -9$, one can get

$$1 - 8 \times 10^{-4} + 2\left(-25 + \frac{\pi}{4}\right) + 40.703 + 1.3\sqrt{2}\frac{\pi}{4} + \left(1.3\sqrt{2}\frac{\pi}{4} + 0.101\right)e^{0.9} < 0 \quad (50)$$

$$1 - 8 \times 10^{-4} + 2\left(-5 + \frac{\pi}{4}\right) + 0.532 + 1.3\sqrt{2}\frac{\pi}{4} + \left(0.11 + 1.3\sqrt{2}\frac{\pi}{4}\right)e^{0.9} < 0 \quad (51)$$

$$1 - 8 \times 10^{-4} - 19.76 + 10.79 + 2.87e^{0.9} < 0. \quad (52)$$

Based on Corollary 2, we know that controlled error neural networks (12) are globally exponentially stable, thus, master neural networks (2) and slave neural networks (7) are globally exponentially synchronized under the designed controller (8). Figs. 8–10 are synchronization errors $e_1(x, t)$, $e_2(x, t)$, and $e_3(x, t)$, respectively. Based on the numerical simulations in Figs. 8–10, it is clear that master neural networks (2) can realize synchronization with slave neural networks (7) under the designed controller (8).

Example 3: Consider neural networks (2) with following parameters: $m = 1$, $i = j = 2$, $a_1 = a_2 = 0.1$, and

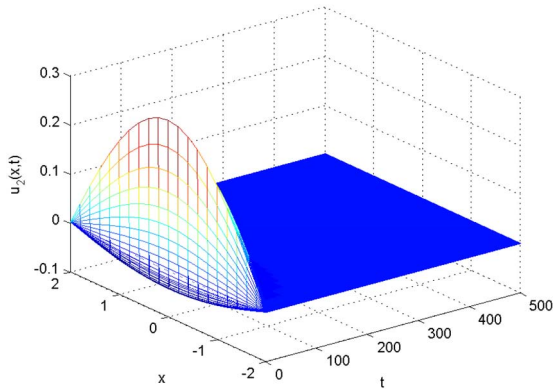


Fig. 12. State trajectories of $u_2(x, t)$ in Example 3.

$$b_1 = b_2 = 0.85$$

$$C = \begin{bmatrix} 0.19 & 0.01 \\ -0.04 & -0.08 \end{bmatrix}, \quad H = \begin{bmatrix} 0.6 & 0.01 \\ -0.05 & 0.5 \end{bmatrix} \quad (53)$$

activation functions $\tilde{f}_i(u) = \tilde{g}_i(u) = \tanh(u)$, $p = 2$, and $x \in \Omega = [-2, 2]$. Hence, we know $l_i = \bar{l}_i = 1$, $i = 1, 2$ and $q_1 = 2$. In this case, [34, Th. 3.1] is invalid. Using [36, Th. 1], the allowable upper bound of the time delay is 2.3. It means that if the time delay $d_{ij}(t)$ is bigger than 2.3, then in [36, Th. 1] is invalid. Yet, utilizing Corollary 3 gives

$$\mathbb{S} = \begin{bmatrix} 0.17 & -0.04 \\ -0.18 & 0.59 \end{bmatrix}. \quad (54)$$

It is clear that the matrix $\mathbb{S}$ is an M -matrix (two eigenvalues are 0.6065 and 0.1535, respectively). According to Corollary 3, one can conclude that neural networks (2) with arbitrary bounded delays are globally exponentially stable. Figs. 11 and 12 are state trajectories of $u_1(x, t)$ and $u_2(x, t)$ with time delay $d_{ij}(t) = 5$, $i, j = 1, 2$, respectively. Figs. 11 and 12 also show the effectiveness of Corollary 3. Thus, the results in this paper are less conservative than the ones in [34] and [36].

V. CONCLUSION

Synchronization is a very interesting collective behavior, which has received much attention in recent years. In this paper, based on theories of PDEs, inequality techniques, and comparison method, algebraic criteria have been developed to ensure master-slave synchronization for a class of reaction–diffusion neural networks with Dirichlet boundary conditions and infinite discrete time-varying delays. Meanwhile, these results were reduced to the ones of finite delays. Finally, efficiency and values of the proposed theoretical criteria were demonstrated by three numerical examples.

Nowadays, stochastic systems have attracted great attention from scientific community because real-world systems are frequently perturbed by environmental noise and human disturbances [67]–[69]. Recall that noise can be utilized to stabilize an unstable system [70]. In the future work, we intend to study dynamical behaviors of stochastic reaction–diffusion neural networks.

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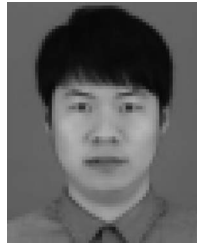
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